

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

12
ELIMINATION OF LEFT RECURSION

DENNIS GUNAWAN

REVIEW

Chomsky Normal Form (CNF):

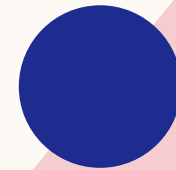
Chomsky Normal Form (CNF)

Conversion of CFG to CNF

OUTLINE

Recursive Production Rules

Elimination of Left Recursion



RECURSIVE PRODUCTION RULES

- Left Recursion

- $A \rightarrow A\beta$

- Examples

- $S \rightarrow Sd$

- $B \rightarrow Bad$

- Right Recursion

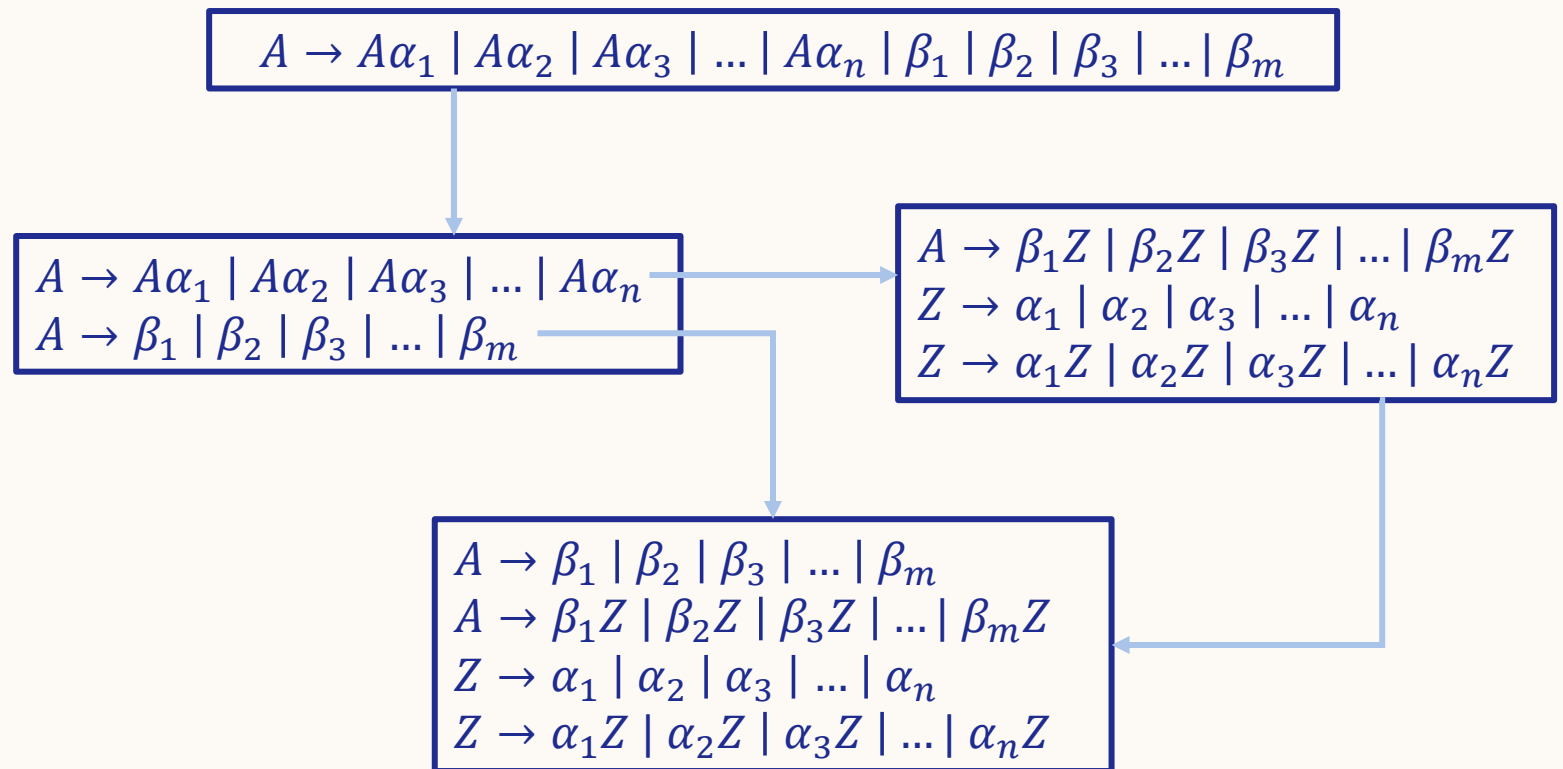
- $A \rightarrow \beta A$

- Examples

- $S \rightarrow dS$

- $B \rightarrow adB$

ELIMINATION OF LEFT RECURSION



ELIMINATION OF LEFT RECURSION

Eliminate left recursion from the following CFG.

$$S \rightarrow \underbrace{Sab}_{\alpha_1} \mid \underbrace{aSc}_{\beta_1} \mid \underbrace{dd}_{\beta_2} \mid \underbrace{ff}_{\beta_3} \mid \underbrace{Sbd}_{\alpha_2}$$

- $S \rightarrow aSc \mid dd \mid ff$
- $S \rightarrow aScZ \mid ddZ \mid ffZ$
- $Z \rightarrow ab \mid bd$
- $Z \rightarrow abZ \mid bdZ$

ELIMINATION OF LEFT RECURSION

Eliminate left recursion from the following CFG.

$$S \rightarrow S \underline{ab} \mid S \underline{b} \mid \underline{cA}$$

$\alpha_1 \quad \alpha_2 \quad \beta_1$

$$A \rightarrow A \underline{a} \mid \underline{a} \mid \underline{bd}$$

$\alpha_1 \quad \beta_1 \quad \beta_2$

- $S \rightarrow cA$
- $S \rightarrow cAZ_1$
- $Z_1 \rightarrow ab \mid b$
- $Z_1 \rightarrow abZ_1 \mid bZ_1$
- $A \rightarrow a \mid bd$
- $A \rightarrow aZ_2 \mid bdZ_2$
- $Z_2 \rightarrow a$
- $Z_2 \rightarrow aZ_2$

ELIMINATION OF LEFT RECURSION

Eliminate left recursion from the following CFG.

$$S \rightarrow S \underline{\alpha_1} a \mid \underline{\alpha_1} a A c \mid \underline{\beta_2} c \mid \underline{\beta_3} \varepsilon$$

$$A \rightarrow A \underline{\alpha_1} b \mid \underline{\beta_1} b a$$

- $S \rightarrow a A c \mid c \mid \varepsilon$
- $S \rightarrow a A c Z_1 \mid c Z_1 \mid Z_1$
- $Z_1 \rightarrow a$
- $Z_1 \rightarrow a Z_1$
- $A \rightarrow b a$
- $A \rightarrow b a Z_2$
- $Z_2 \rightarrow b$
- $Z_2 \rightarrow b Z_2$



PRACTICE

EXERCISES

1. Eliminate left recursion from the following CFG.

$$A \rightarrow Aa \mid aBc$$

2. Eliminate left recursion from the following CFG.

$$\begin{aligned} A &\rightarrow AbAB \mid \varepsilon \\ B &\rightarrow BAa \mid A \mid \varepsilon \end{aligned}$$

EXERCISES

3. Eliminate left recursion from the following CFG.

$$S \rightarrow SBa \mid Ab$$

$$A \rightarrow Sa \mid AAb \mid a$$

$$B \rightarrow Sb \mid BBa \mid b$$

4. Eliminate left recursion from the following CFG.

$$S \rightarrow SSC \mid SSB \mid abg$$

$$B \rightarrow abc \mid BSb \mid BCd$$

$$C \rightarrow ab$$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

Greibach Normal Form (GNF):

Greibach Normal Form (GNF)

Conversion of CNF to GNF

VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



iF INFORMATIKA
UMN